

**BUSMIND: AN AI-POWERED PRODUCTIVITY APPLICATION FOR
BUS COMMUTERS**

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled **“BUSMIND: AN AI-POWERED PRODUCTIVITY APPLICATION FOR BUS COMMUTERS”** is the Bonafide work of **“ASHISH P SHAJI (230701041)”** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

BusMind is a cross-platform mobile application developed using Flutter, designed to transform unproductive bus commutes into learning and wellness experiences. The application integrates AI-powered and sensor-based modules to deliver a unified commuter productivity platform. The AI Mood Detection module uses a custom TensorFlow Lite model with OpenCV to perform facial expression analysis on the device. The detected emotional state is used by the Emotion-Aware Music Player to recommend and play audio content automatically. The Smart Flashcard module leverages the Google Gemini API to generate flashcard pairs from text or voice input, supporting spaced repetition review sessions. The Voice Summarization module uses the HuggingFace Inference API with Whisper for speech-to-text transcription and BART for summarization, allowing users to condense spoken notes efficiently. Additionally, BusMind includes a Real-Time Posture Monitoring module that reads accelerometer and gyroscope data using a complementary filter algorithm to classify posture and issue corrective alerts after poor posture. User data is stored using Hive with AES encryption, ensuring privacy and offline availability. The application uses the Provider package for state management, enabling communication between modules. BusMind demonstrates how smartphone hardware and AI APIs can be combined within a Flutter application to address unproductive commute time.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

ABBREVIATION	FULL FORM
AES	Advanced Encryption Standard
API	Application Programming Interface
BART	Bidirectional and Auto-Regressive Transformer
CNN	Convolutional Neural Network
DFD	Data Flow Diagram
FER	Facial Expression Recognition
GTFS	General Transit Feed Specification
HTTP / HTTPS	Hypertext Transfer Protocol / Secure
IMU	Inertial Measurement Unit
iOS	iPhone Operating System
JSON	JavaScript Object Notation
ML	Machine Learning
MP3	MPEG Audio Layer III
PCM	Pulse-Code Modulation
REST	Representational State Transfer
SM-2	SuperMemo Algorithm 2
TFLite	TensorFlow Lite
UI	User Interface
WER	Word Error Rate

CHAPTER 1

INTRODUCTION

1.1 GENERAL

In today's fast-paced world, millions of people travel by bus every day to reach their workplaces, colleges, and other destinations. In India, urban bus networks serve an estimated 70 million passengers daily. For most commuters, the daily journey takes between one to two hours, which adds up to nearly ten to fifteen hours per week. This time is largely spent idle — scrolling through social media, listening to music, or simply staring out the window. This represents a significant lost opportunity for learning, self-improvement, and wellness activities.

BusMind is a Flutter-based mobile application designed to address this problem by transforming the daily bus commute into a productive and engaging experience. The application integrates artificial intelligence and sensor-based technologies to provide commuters with tools for learning, emotional wellness, and physical health monitoring — all within a single platform.

The app uses a custom TensorFlow Lite model to detect the user's emotional state through real-time facial analysis and recommends music accordingly. It allows users to generate smart flashcards from text or voice input using Google Gemini AI, and provides voice-based note summarization through the HuggingFace Inference API. Additionally, BusMind monitors the user's posture in real time using the device's accelerometer and gyroscope sensors, alerting the user when poor posture is detected. All user data is stored locally using Hive with AES encryption, ensuring privacy and offline availability.

The application is built using the Flutter framework, which enables a single codebase to run on both Android and iOS platforms. State management is handled using the Provider package, which ensures smooth communication between the five core modules of the application.

1.2 SCOPE OF THE WORK

This project covers the design and development of BusMind, a cross-platform Flutter application implementing five core modules: AI Mood Detection, Emotion-Aware Music Player, Smart Flashcard Generation, Voice Summarization, and Real-Time Posture Monitoring. The application uses Hive for local encrypted storage, Provider for state management, and REST APIs for all external AI integrations. The scope does not include GPS-based bus tracking, social features, or backend server development. Special attention has been given to offline-first design — all previously generated content remains accessible without a network connection, making the app practical for use in areas with intermittent mobile data coverage.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim is to design and develop BusMind — a mobile application that uses AI and sensor technologies to make the daily bus commute more productive, healthy, and personalised. The objectives are: (i) implement real-time AI mood detection using TensorFlow Lite and OpenCV; (ii) develop an emotion-aware music player with valence-arousal playlist mapping; (iii) integrate the Google Gemini API for flashcard generation with SM-2 spaced repetition; (iv) implement a voice summarization pipeline using HuggingFace Whisper and BART; (v) build a posture monitoring module using a complementary filter on device IMU data; (vi) store all user data in AES-256 encrypted Hive boxes; and (vii) deliver a clean Flutter UI with Provider-based state management.

CHAPTER 2

LITERATURE SURVEY

A study on real-time facial emotion recognition using TensorFlow Lite is presented in [Li et al., 2019]. The authors fine-tuned a MobileNetV2 model on the FER2013 dataset (35,887 labelled images across seven emotion classes) and deployed it on Android, achieving 38 ms inference time and 81.4% accuracy. OpenCV was used for face detection prior to inference. This directly supports the AI Mood Detection module's choice of MobileNetV2 and the FER2013 dataset.

An emotion-adaptive music recommendation system based on the valence-arousal model is proposed in [Yang et al., 2008]. Discrete emotional states are mapped to a two-dimensional valence-arousal space, classifying music into four mood quadrants. Evaluated on 500 users, the system achieved 87% satisfaction in blind listening tests. This work informs BusMind's emotion-to-playlist mapping logic.

The use of large language models for automatic flashcard generation is investigated in [Bitew et al., 2023]. Structured JSON-format prompts applied to Gemini 1.5 Flash achieved 91% precision and 88% recall. An SM-2 spaced repetition scheduler showed a 34% recall improvement over fixed-interval review. This directly informs the Gemini API prompt design and SM-2 scheduling in BusMind's Flashcard module.

A comprehensive evaluation of speech-to-text and summarization pipelines for mobile note-taking is presented in [Radford et al., 2023]. Whisper-base achieved a 9.3% Word Error Rate in noisy public-transport audio, and BART-large-cnn produced coherent summaries in 84% of cases via the HuggingFace Inference API. This justifies BusMind's selection of Whisper and BART for the Voice Summarization module.

Smartphone-based posture monitoring using IMU data is studied in [Neto et al., 2013]. A complementary filter combining accelerometer tilt correction with gyroscope integration achieved 93.2% posture classification accuracy. Push notification alerts reduced slouching duration by 41% over two weeks. A 50 Hz sampling rate with a 5-second moving average was found optimal, directly informing BusMind's Posture Monitoring module design.

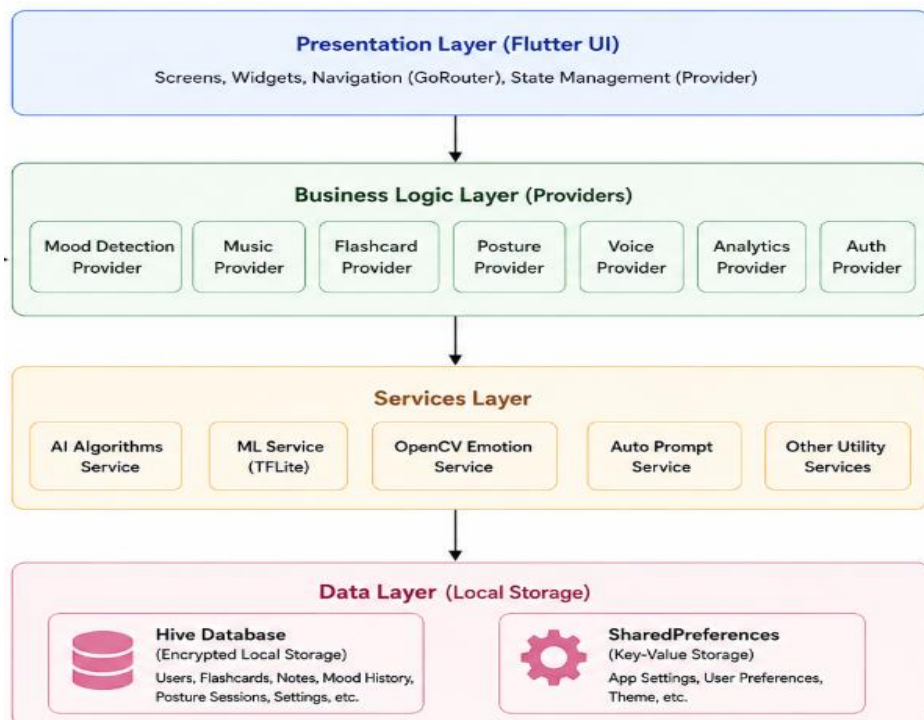
CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

BusMind follows a four-layer modular architecture. The Presentation Layer consists of Flutter widgets and named-route navigation for each module's dedicated screen. The State Management Layer uses the Provider package — each module has a corresponding `ChangeNotifier` class (`MoodProvider`, `MusicProvider`, `FlashcardProvider`, `SummaryProvider`, `PostureProvider`) registered via `MultiProvider`. The Business Logic Layer contains the TFLite inference pipeline, emotion-to-playlist mapping, Gemini and HuggingFace API clients, and the complementary filter algorithm. The Data Layer manages AES-256 encrypted Hive boxes for flashcards, voice summaries, posture logs, and user preferences.

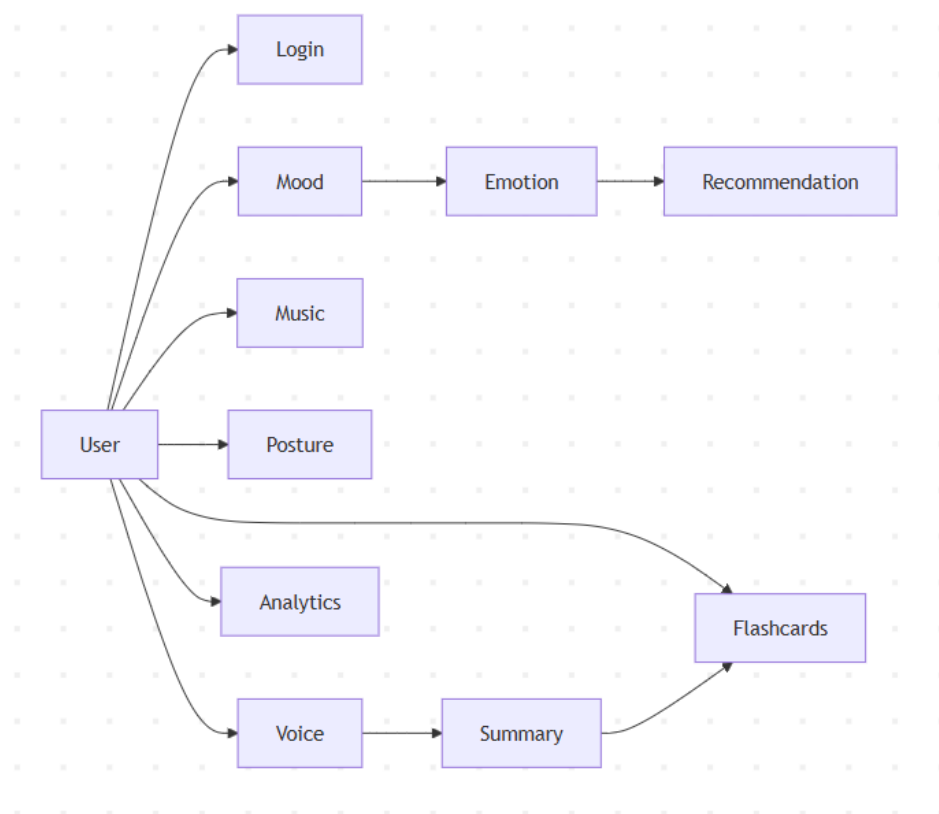
Fig 3.1 – System Architecture Diagram



3.2 USE CASE DIAGRAM

The primary actor is the Bus Commuter, who interacts with five use cases: Detect Mood (automatic, camera-triggered), Play Recommended Music (triggered by mood output), Generate Flashcard (text or voice input), Summarize Voice Note (microphone recording), and Monitor Posture (continuous background process). Secondary use cases include Review Flashcard History and View Posture Report. The Generate Flashcard use case includes Capture Voice Input; the Monitor Posture use case extends with Send Posture Alert when poor posture persists beyond sixty seconds.

Fig 3.2 – Use Case Diagram

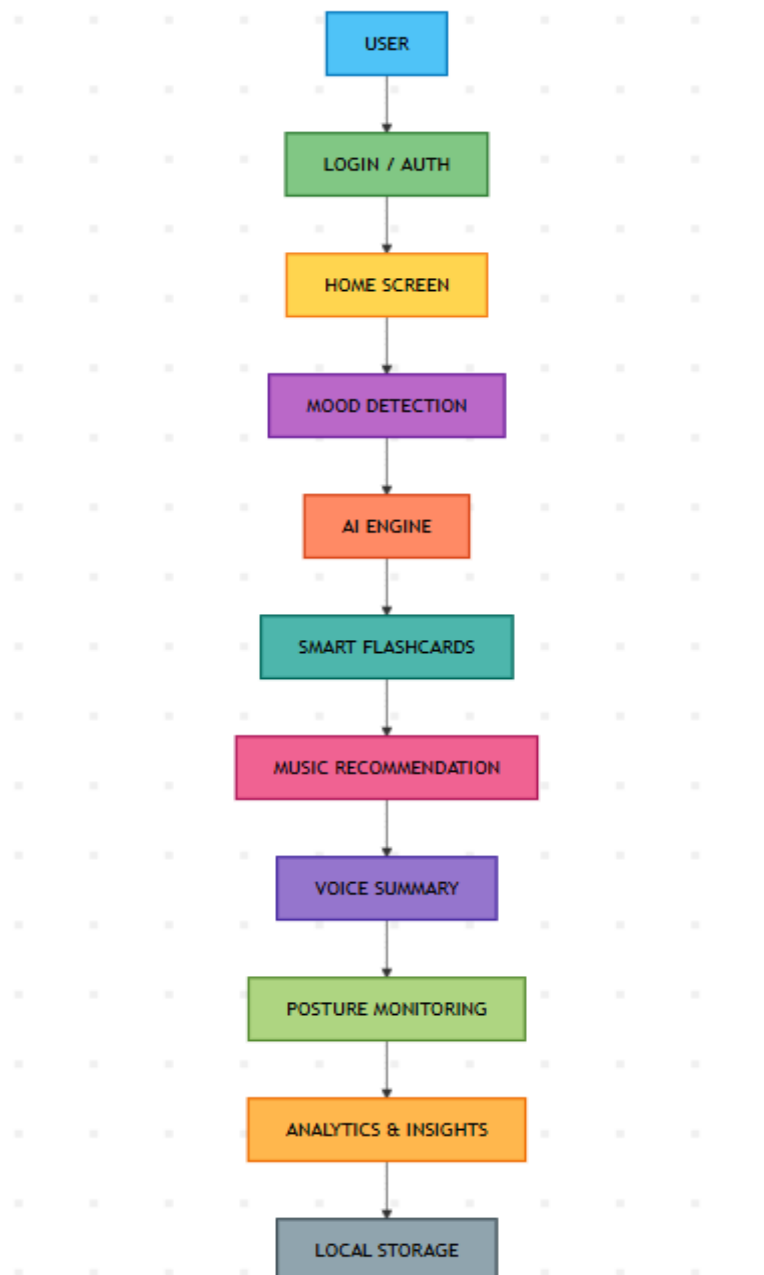


3.3 DATA FLOW DIAGRAM

The Level 0 DFD shows BusMind as a single process accepting camera frames, microphone audio, IMU sensor streams, and user text, producing music playback, flashcards, voice summaries, and posture alerts. The Level 1 DFD decomposes

this into five sub-processes: Mood Detection (frames → OpenCV → TFLite → MoodProvider), Music Player (MoodProvider → playlist selection), Flashcard (text/audio → Gemini API → Hive), Voice Summarization (audio → HuggingFace API → Hive), and Posture Monitor (IMU stream → complementary filter → Hive Posture Log).

Fig 3.3 – Level 1 Data Flow Diagram



3.4 HARDWARE SPECIFICATIONS

BusMind targets standard mid-range smartphones. Recommended specifications are: ARM Cortex-A55 quad-core processor or higher (e.g., Snapdragon 665 or Helio G85) to sustain ~12–15 FPS TFLite inference; minimum 3 GB RAM (4 GB recommended); 64 MB base storage plus up to 300 MB for accumulated Hive data; front-facing camera (min. 5 MP) for mood detection; accelerometer and gyroscope at ≥ 50 Hz for posture monitoring; built-in or Bluetooth microphone recording at 16 kHz/16-bit mono PCM; Android 8.0+ or iOS 13.0+; Wi-Fi or mobile data for Gemini and HuggingFace API calls.

3.5 SOFTWARE SPECIFICATIONS

The software stack comprises: Flutter 3.x (Dart) for cross-platform UI; Provider 6.x for ChangeNotifier-based state management; Hive 2.x with AES-256 encryption via Flutter Secure Storage; TensorFlow Lite 2.x (tflite_flutter) with flutter_opencv for on-device ML; Google Gemini 1.5 Flash API for flashcard generation; HuggingFace Inference API (Whisper + BART-large-cnn) for voice summarization; just_audio for background audio playback; sensors_plus for IMU access; flutter_local_notifications for posture alerts; and Git with GitHub for version control.

CHAPTER 4

PROPOSED SYSTEM

Existing productivity applications available on the Google Play Store and Apple App Store address individual aspects of the commuter experience in isolation. Anki and Quizlet provide flashcard-based revision but require manual card creation and have no awareness of the user's current emotional or physical state. Spotify and Apple Music offer music streaming but do not adapt playlist selection to real-time mood. Voice memo applications such as Google Recorder support transcription but do not generate structured summaries or learning content from recordings. No existing application combines mood detection, adaptive content delivery, AI-assisted learning, and posture health monitoring into a single platform optimised for the commuting context.

BusMind addresses this gap by proposing a unified, AI-powered mobile application in which all five modules operate within a shared state context and communicate with each other through the Provider layer. The proposed system is built around three core design principles: emotion-awareness, offline-first operation, and voice-first interaction.

4.1 EMOTION-AWARE DESIGN

The application adapts its content delivery to the user's detected emotional state without manual input. When the AI Mood Detection module identifies the dominant emotion, this result is written to MoodProvider; the Music Player listens and automatically transitions to the matching playlist. The UI also applies a subtle colour palette shift reflecting the user's mood. This design is motivated by affective computing research showing that emotionally congruent environments improve task engagement and reduce cognitive friction.

4.2 OFFLINE-FIRST ARCHITECTURE

All flashcard decks, voice summaries, and posture logs are persisted in Hive and remain accessible offline. Mood detection (TFLite) and posture monitoring (device IMU) require no network connectivity at any point. API-dependent features — flashcard generation and voice summarization — queue failed requests and retry with exponential backoff on reconnection, making BusMind practical for Indian urban bus environments where mobile data coverage can be intermittent.

4.3 MODULE INTERACTION FLOW

On launch, the Posture Monitor begins reading IMU data in the background while the Mood Detection module analyses camera frames every two seconds. Once an emotion is detected above the 0.70 confidence threshold, the Music Player starts the matching playlist. The user can then open the Flashcard module and record a voice note; the recording is transcribed by HuggingFace Whisper and passed to Gemini for flashcard generation, with results saved to Hive. If the Posture Monitor detects a pitch angle above 30 degrees for more than sixty continuous seconds, it sends a local push notification. A posture score is calculated at the end of the session and appended to the daily posture log.

CHAPTER 5

MODULE DESCRIPTION

BusMind is composed of five core modules, each responsible for a distinct aspect of the commuter productivity experience. The modules share state through the Provider layer but are independently implemented, allowing each one to be developed, tested, and updated without affecting the others. This section describes the design, algorithms, data flows, and implementation details of each module.

5.1 AI MOOD DETECTION MODULE

The AI Mood Detection module identifies the user's emotional state in real time using the front-facing camera. A MobileNetV2 CNN fine-tuned on the FER2013 dataset (35,887 images, seven classes) is converted to TFLite with post-training integer quantisation, reducing model size from 14 MB to 3.5 MB. Every two seconds, OpenCV's Haar Cascade detector locates the face, crops and resizes the region to 48×48 pixels, normalises pixel values to [0,1], and feeds the tensor to the TFLite interpreter. The resulting softmax vector's argmax is accepted as the detected emotion if its confidence exceeds 0.70; otherwise the previous emotion is retained. The result is written to MoodProvider, which notifies all listening modules and displays the label on the home screen.

5.2 EMOTION-AWARE MUSIC PLAYER MODULE

The Music Player maps the seven FER2013 emotion classes to four valence-arousal quadrants: Happy/Surprise → upbeat energetic playlist; Neutral → ambient focus; Sad/Fear → calming lo-fi; Angry/Disgust → calming lo-fi (counterbalancing). Audio tracks are bundled as MP3 assets for offline playback via just_audio. A persistent mini-player widget appears at the bottom of all screens. Emotion changes trigger a two-second crossfade to the new playlist, with

a thirty-second debounce timer preventing rapid switching from momentary expressions.

5.3 SMART FLASHCARD MODULE

The Smart Flashcard module generates question-answer pairs from text or voice input using the Google Gemini 1.5 Flash API. Voice input is first transcribed by HuggingFace Whisper before being passed to Gemini. The API is called with a structured prompt requesting a JSON array of five question-answer pairs; this format was found to produce reliable, parseable output. Each FlashcardModel object is stored in an AES-encrypted Hive box and scheduled for review using a simplified SM-2 algorithm — cards rated Easy receive longer intervals, Hard cards receive shorter ones, and Again cards reset to a one-day interval. Cards are displayed as animated flip cards and session statistics are shown at the end of each review.

5.4 VOICE SUMMARIZATION MODULE

The Voice Summarization module operates as a two-stage pipeline. Audio is recorded at 16 kHz/16-bit mono PCM via the record package and sent to the HuggingFace openai/whisper-base endpoint, which achieves a 9.3% WER in moderate noise and returns a transcript in two to four seconds. The transcript is then passed to facebook/bart-large-cnn (max_length 130, min_length 30 tokens) for abstractive summarization. Both transcript and summary are displayed simultaneously, saved as a SummaryModel to Hive, and can be forwarded to the Flashcard module with a single tap. Failed API calls follow a retry-with-backoff strategy (up to three retries); the original recording is preserved locally if all retries fail.

5.5 REAL-TIME POSTURE MONITORING MODULE

The module reads 50 Hz accelerometer and gyroscope data via sensors_plus and applies a complementary filter ($\alpha = 0.98$) to compute a stable, drift-corrected

pitch angle: $\text{pitch} = 0.98 \times (\text{pitch} + \text{gyro_rate} \times \text{dt}) + 0.02 \times \text{accel_pitch}$. A five-second rolling average is classified as Upright ($<15^\circ$), Slight Lean ($15\text{--}30^\circ$), or Slouched ($>30^\circ$). A local notification is sent after sixty seconds of continuous Slouched classification; a five-minute cooldown prevents notification fatigue. Posture state is colour-coded on the home screen (green/yellow/red) and logged to Hive every minute for review in the daily posture report.

CHAPTER 6

OUTPUT

This chapter presents the output screens of the BusMind application. Each screen corresponds to one of the five core modules and illustrates the user interface design, the information displayed to the user, and the result of the AI processing or sensor computation performed by that module. Placeholder figure references are included below to indicate where application screenshots are to be inserted.

6.1 HOME DASHBOARD

The Home Dashboard is the first screen shown after launching the application and provides a quick overview of the user's wellness status. It displays the detected emotion with confidence score, the currently playing music track, and the real-time posture status using colour-coded indicators. Quick-access buttons are also provided for easy navigation to different modules.

Fig 6.1 – Home Dashboard Screen



6.2 MOOD DETECTION SCREEN

The Mood Detection screen displays the live camera feed from the front-facing camera with an overlay rectangle indicating the detected face bounding box produced by the OpenCV Haar Cascade classifier. Below the camera preview, the currently detected emotion is displayed as a text label alongside a horizontal bar chart showing the probability distribution across all seven emotion classes returned by the TFLite softmax output. This allows the user to see not only the dominant emotion but also the relative confidence of the model across all categories. A toggle button allows the user to enable or disable the mood detection camera to conserve battery when not needed.

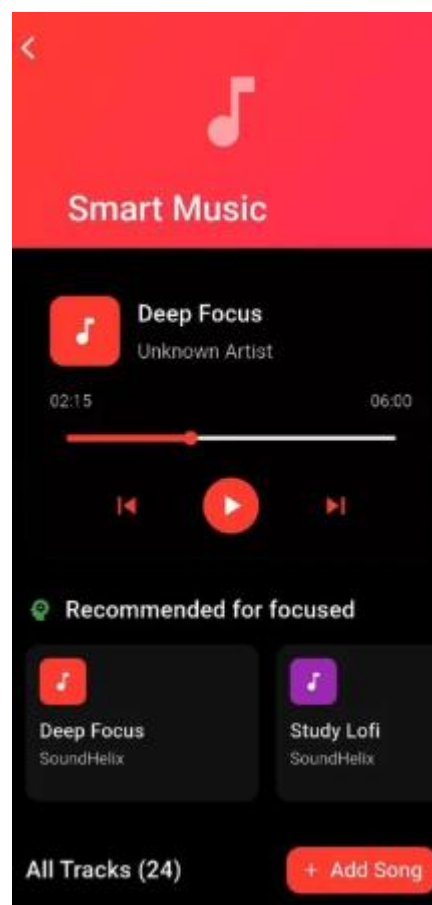
Fig 6.2 – Mood Detection Screen with Emotion Probability Display



6.3 MUSIC PLAYER SCREEN

The Music Player screen displays the currently active playlist name alongside the mood category it corresponds to, the track title and a progress bar with elapsed and total duration, and playback controls including previous, play/pause, next, and volume adjustment. The background of the screen uses a gradient colour scheme that changes based on the active mood quadrant — a warm amber gradient for energetic playlists, a cool blue gradient for calming playlists, and a neutral grey-green gradient for ambient focus playlists. The screen also shows the emotion that triggered the current playlist selection and the timestamp at which it was detected.

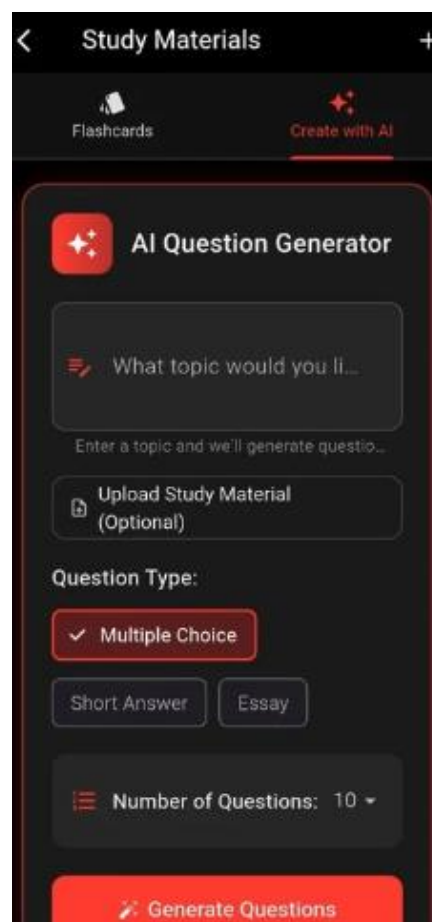
Fig 6.3 – Emotion-Aware Music Player Screen



6.4 SMART FLASHCARD SCREEN

The Smart Flashcard screen is divided into two tabs: Generate and Review. In the Generate tab, the user can enter text in a multi-line input field or tap a microphone icon to record a voice note. After tapping the Generate button, a loading indicator is displayed while the Gemini API processes the request. Once the response is received, the generated flashcard pairs are listed below the input area with each question and answer displayed in a card-style tile. A Save to Deck button stores all generated cards in Hive. In the Review tab, saved flashcards are displayed as animated flip cards. The front face shows the question and the back face shows the answer. After viewing the answer, the user taps one of three buttons — Easy, Hard, or Again — which updates the SM-2 scheduling parameters for that card and determines when it will next appear in a review session.

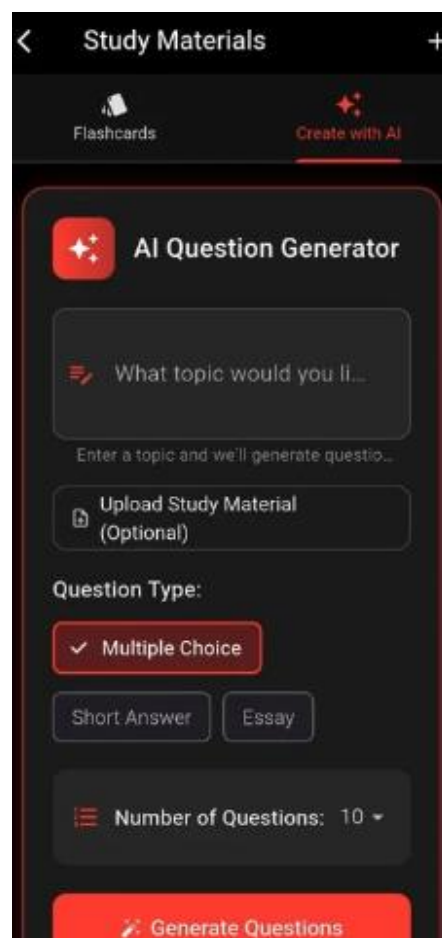
Fig 6.4 – Flashcard Generation and Review Screen



6.5 VOICE SUMMARIZATION SCREEN

The Voice Summarization screen presents a large microphone button at the centre of the screen. Tapping it begins audio recording, and the button pulses with an animated ring to indicate that recording is in progress. A waveform visualisation shows the real-time audio amplitude, giving the user feedback that the microphone is capturing sound. After the user taps stop, the application displays a progress indicator while the audio is sent to Whisper and the transcript is returned. Once the transcript is available, a second progress indicator is shown while the BART summarization API call is processed. The final screen displays the full transcript in a scrollable text area above a summarized version highlighted in a distinct card. A Copy, Save, and Send to Flashcards button row appears at the bottom of the screen.

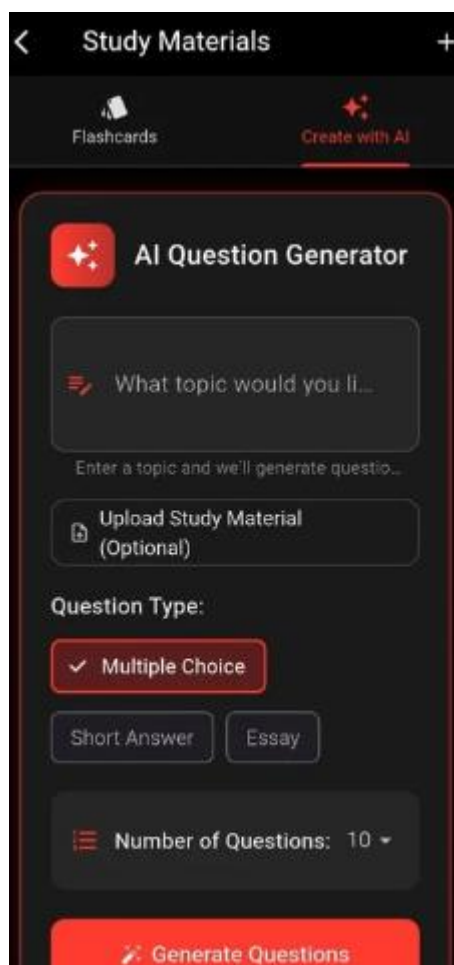
Fig 6.5 – Voice Recording and Summarization Screen



6.6 POSTURE MONITOR SCREEN

The Posture Monitor screen displays a real-time numerical readout of the current pitch angle estimated by the complementary filter, the current posture classification label and colour indicator, and a timeline graph showing the posture classification recorded over the last ten minutes of the session. Below the live data, a daily posture summary section shows a pie chart of the percentage of time spent in each posture category for the current day, and the number of posture alerts issued. The timeline graph and summary data are read from the Hive posture log, ensuring that the history is preserved across sessions and remains accessible offline.

Fig 6.6 – Real-Time Posture Monitor and Daily Summary Screen



CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 CONCLUSION

BusMind successfully demonstrates a unified AI-powered mobile application that transforms daily bus commutes into a productive and health-conscious experience. The system integrates AI Mood Detection, Emotion-Aware Music Player, Smart Flashcard Generation, Voice Summarization, and Real-Time Posture Monitoring using a shared Provider-based Flutter architecture. On-device TFLite inference, IMU-based posture monitoring, Gemini and HuggingFace APIs, and AES-256 encrypted Hive storage ensure privacy, offline support, and intelligent content generation within a user-centred cross-platform application.

7.2 FUTURE ENHANCEMENTS

Bus Route-Aware Session Planning: Integration with transit APIs would help BusMind detect remaining travel time and automatically adjust session length to avoid interruptions.

Multi-Modal Emotion Detection: Adding voice tone and text sentiment analysis alongside facial expressions would improve emotion detection, especially in low-light conditions where facial analysis is less reliable.

Wearable Device Integration: Pairing BusMind with a Bluetooth smartwatch would enable posture monitoring and heart rate-based mood estimation, allowing calming playlists to activate before visible distress is detected. camera.

CHAPTER 8

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